Empirical Validation Architecture for

Cognitive State Research

CSRVA — Version 5.1

Dimensionally Consistent Measurement Protocols, Falsifiable Predictions, and Ethical Safeguards for the Computational Framework of Altered Social Cognition

Compliance Domain	Status	Standard
Dimensional consistency	Enforced throughout	SI units; verify with dimensional analysis before use
Statistical rigour	Pre-registration required	OSF.io or ClinicalTrials.gov before data collection
Literature grounding	≥2 citations per claim	Primary sources only; no secondary citation chains
Clinical safeguards	Mandatory	APA Ethical Principles; IRB approval required
Falsifiability	Null conditions defined for all equations	Every claim specifies a falsifying outcome
Code reproducibility	Python pseudocode with full contracts	Containerise before deployment; DOI on Zenodo
Scope limitation	Enforced	No paranormal claims; all phenomena within known physics

HONEST CLASSIFICATION: This protocol describes the scientific work required to validate the accompanying computational philosophy framework. The protocol itself is not data. It is a research design.

CC BY-NC-SA 4.0 | Research design only — not clinical guidance

PREAMBLE

From Philosophy to Science

The five-step transition this protocol enables

The accompanying Computational Framework for Altered Social Cognition makes numerous claims about relationships between measurable neural parameters and subjective cognitive experience. Those claims are currently computational philosophy — precise, internally consistent, and empirically unverified.

This document specifies what would need to happen for those claims to become science. It is organised around the five conditions identified in the framework's preface.

Step	Requirement	This Document's Role
1	Isolate a specific, falsifiable hypothesis	Section 3 provides 6 complete pre-registerable experimental designs
2	Pre-register before data collection	Section 3 includes registration template elements
3	Secure funding, ethics approval, and informed consent	Section 5 provides IRB protocol elements and consent language
4	Collect data, analyse, publish (including nulls)	Section 4 provides validated analysis pipelines
5	Independent replication	Section 6 provides reporting standards for replication

[IMPORTANT] This protocol contains no data. Its honest status is: a research design for hypotheses that have not yet been tested. A detailed research design is the necessary precondition for science — but it is not itself science.

SECTION 1

Operationalised Measurement Axes

P-B-T with explicit units, measurement protocols, and null conditions

The P-B-T framework's three axes must each have a defined measurement protocol before any hypothesis can be tested. The following specifications translate abstract computational constructs into measurable, reportable quantities.

1.1 Precision Axis (P)

Definition Signal weighting relative to prior expectations; indexed by brain's gain on incoming evidence

Units $\mu V \cdot ms^{-1}$ (ERP amplitude per unit reaction time standard deviation)

```
P = (A_MMN / σ_noise) · (1 / τ_RT)
```

A_MMN = mismatch negativity peak amplitude (µV) in 100–250ms post-deviant window; σ_noise = RMS noise in 200ms pre-stimulus baseline; τ_RT = within-session reaction time SD (ms)

Equipment 128-channel EEG; ≥500 Hz sampling; impedance < 10 kΩ

Paradigm Auditory oddball: 80% standard (1000 Hz) / 20% deviant (1100 Hz); ≥200 deviant stimuli; jittered ITI 800–1200 ms

Source Näätänen et al. (2007) Clin Neurophysiol 118:2544; Friston (2005) Philos Trans R Soc B 360:1023

Null condition P_stimulant − P_placebo = 0 at FDA-approved single dose; test with double-blind crossover RCT

1.2 Boundary Axis (B)

Definition Functional separation between self-referential (DMN) and task-directed processing

Units Dimensionless ratio of Fisher-z transformed Pearson correlation coefficients

```
B = FC_DMN-internal / FC_DMN-external
```

FC_DMN-internal = mean correlation between DMN nodes (posterior cingulate cortex, medial PFC, angular gyrus); FC_DMN-external = mean correlation between DMN nodes and salience/frontoparietal network nodes

Equipment 3T fMRI minimum; TR ≤ 2s; 8-minute resting-state acquisition; exclude volumes with > 0.3 mm framewise displacement

Parcellation Yeo 17-network atlas (2011, J Neurophysiol 106:1125)

Source Fox et al. (2005) PNAS 102:9673; Andrews-Hanna et al. (2014) Neuron 84:269

Null condition B does not change after saline vs. intranasal oxytocin 24 IU; double-blind RCT, N=60

1.3 Temporal Axis (T)

Definition Future vs. past orientation of reward prediction; indexed by delay discounting parameter k

Units Dimensionless z-score relative to matched population norms

```
T = log(k_individual / k_population_median)   z-scored against matched normative sample
```

k from hyperbolic discounting V = A / (1 + k·D); Kirby & Maraković 27-item task; k in units of 1/days; normative sample: age, sex, education matched, N ≥ 200

Source Mazur (1987) Quantitative Analyses of Behavior Vol. 5; Bickel et al. (2014) Annu Rev Clin Psychol 10:1

Null condition T does not predict accuracy above 50% in a pre-registered forced-choice social inference task

1.4 Polytope Bounds — Population Quantile Method

The original framework used arbitrary numerical thresholds. These must be replaced with empirically derived population quantile estimates from a normative sample before testing clinical or experimental groups.

```
σ_max = Q_0.95(σ_healthy)     ρ_max = Q_0.95(ρ_healthy)     η_max =
Q_0.95(η_r,healthy)
```

Q_0.95 = 95th percentile of the healthy population distribution; estimated from N ≥ 500 normative sample prior to experimental use

Published estimates σ_max ≈ 8.2 µV (Winterer et al., 2004, Neuropsychopharmacology 29:1170); ρ_max ≈ 0.94 (Stam et al., 2005); η_max ≈ 0.88 (Müller et al., 2005, Neurosci Lett 373:119)

⚠️ **These are estimates from studies not designed to measure P-B-T. Re-estimate in your specific measurement context before applying thresholds.**

SECTION 2

Grounded Equations

Dimensionally verified replacements for speculative versions

Each equation below is either taken directly from established literature [GROUNDED] or is a novel hypothesis [HYPOTHESIS] with units verified, derivation cited, and null condition specified.

Eq 1 — Precision Dynamics [HYPOTHESIS]

$$dP/dt = \alpha\cdot([DA]/[DA]_0)\cdot(\delta^2/\sigma_y^2) - (P-P_0)/\tau_P + \xi(t)$$

Units: $\mu V\cdot ms^{-1}\cdot s^{-1}$ | derivation: homeostatic relaxation + dopaminergic gain; follows Friston (2009) Neuroimage 48:513

Null: dP/dt = 0 at rest without dopaminergic perturbation

Eq 2 — Boundary Dynamics [HYPOTHESIS]

$$dB/dt = -(B - B_0)/\tau_B + \kappa \cdot \Delta H_arousal / (k_B \cdot T_body)$$

Units: dimensionless$\cdot s^{-1}$ | $k_B\cdot T_body = 4.14\times10^{-21}$ J at 310K renders the perturbation dimensionless

Null: B does not change after sham TMS to angular gyrus

Eq 3 — Spectral Radius [GROUNDED]

$$\rho(W) = \max_i\ |\lambda_i(W)| \quad W = \text{MVAR coefficient matrix from EEG}$$

Units: dimensionless | MVAR order: AIC-selected; stationarity: ADF test required

Stam et al. (2005) Clin Neurophysiol 116:2266; Rajan & Abbott (2006) Phys Rev Lett 97:188104

Eq 4 — Phase-Locking Value [GROUNDED]

$$PLV_AB(f) = |\ 1/T \cdot \Sigma_t \exp(i\cdot(\varphi_A(t,f) - \varphi_B(t,f)))\ |$$

Units: dimensionless [0,1] | volume conduction corrected: subtract mean of 200 phase-randomised surrogates

Lachaux et al. (1999) Hum Brain Mapp 8:194

Eq 5 — Mutual Information [GROUNDED]

$$I(A;B) = H(A) + H(B) - H(A,B) \quad [\text{KSG k-NN estimator, } k=3]$$

Kraskov et al. (2004) Phys Rev E 69:066138; bias corrected via 200 phase-shuffled surrogates

Eq 6 — Free Energy (Active Inference) [GROUNDED]

$$F = D_KL[q(z|x)\ ||\ p(z)] + E_\{q(z|x)\}[-\ln p(x|z)]$$

Friston (2010) Nat Rev Neurosci 11:127; complexity cost + inaccuracy cost

Eq 7 — Cognitive State Prediction Function [HYPOTHESIS]

$$\Psi_mind(t) = F(P(t), B(t), T(t), W(t), S(t)) + \varepsilon(t)$$

F estimated from data; $\varepsilon(t) \sim N(0, \sigma_\varepsilon^2)$ estimated via cross-validation

Acceptance criterion: cross-validated $R^2 > 0.15$; full model improves over best single-axis model by $\Delta R^2 > 0.05$

Eq 8 — Critical Slowing Down Indicator [GROUNDED]

$$\varphi_1(t) = \mathrm{Cov}(x_t, x_{t-1}) / \mathrm{Var}(x_t) \quad \text{[AR(1) in sliding 30s window]}$$

Scheffer et al. (2009) Nature 461:53 — $\varphi_1 \to 1$ as system approaches bifurcation

Prediction: $\varphi_1 > 0.85$ in the 12 hours preceding clinically confirmed manic episode onset

Eq 9 — Granger Causality [GROUNDED]

$$GC(X \to Y) = \ln(\mathrm{Var}(\varepsilon_{Y|Y\text{-past}}) / \mathrm{Var}(\varepsilon_{Y|Y\text{-past},X\text{-past}}))$$

Granger (1969) Econometrica 37:424; Bressler & Seth (2011) Neuroimage 58:323

Application: test whether stimulant-induced ΔP Granger-causes ΔB; lag via AIC; ADF stationarity test required

Eq 10 — Allostatic Load [GROUNDED]

$$AL = \Sigma_i z_i^2 \quad z_i = (x_i - \mu_i) / \sigma_i$$

McEwen & Seeman (1999) Ann NY Acad Sci 896:30; biomarkers: cortisol, BP, HR, BMI, HbA1c, CRP

SECTION 3

Pre-Registerable Predictions

6 complete experimental designs

Each prediction below translates a framework claim into a complete, pre-registerable experimental design. None have been conducted. Each requires IRB approval and pre-registration before any data collection.

Prediction 1 — P and B jointly predict social inference accuracy

Hypothesis Higher P and lower B predict above-chance accuracy on a standardised social inference task, over and above verbal IQ

Primary DV Proportion correct on 60-trial video-based emotion recognition task (Ekman micro-expression stimuli, 1-second clips)

Primary IVs P (continuous, EEG MMN protocol), B (continuous, resting fMRI), P×B interaction; covariate: WASI verbal IQ

Control condition Same session: scrambled-face stimuli matched for visual complexity

Required N N=96 (G*Power: multiple regression, f^2=0.15, power=0.80, α=0.05, k=3 predictors)

Analysis Multiple linear regression; Holm–Bonferroni correction; report β, SE, 95% CI for each predictor

Null condition Prediction fails if $\beta_P \leq 0$, OR $\beta_B \geq 0$, OR model F-test $p > 0.05$ after covariate control

Prediction 2 — Spectral radius ρ ≈ 1 during self-reported flow states

Hypothesis ρ from MVAR-EEG is closer to 1.0 during self-reported flow than during boredom or anxiety, within subjects

Primary DV Spectral radius ρ (per 30-second window)

Primary IV Self-reported state: flow vs. boredom vs. anxiety (3-level, within-subjects)

Task Musical instrument or video game performance; Flow Short Scale immediately post-block; interruption condition as flow-prevention control

Required N N=60 (LMM power simulation: ICC=0.40, medium effect, power=0.80)

Analysis Linear mixed model: ρ ~ State + (1|Subject); pairwise contrasts flow vs. boredom and flow vs. anxiety; Δρ with 95% CI

Null condition Prediction fails if $\rho_flow - \rho_boredom$ CI includes 0

Prediction 3 — Gamma PLV precedes accurate empathy judgements

Hypothesis Surrogate-corrected inter-brain gamma-band PLV (30–80 Hz) is higher in the 200ms before accurate emotion recognition trials than before inaccurate trials

Confound controls LCMV beamforming; eye movement rejection; equated trial counts for accuracy levels

Required N N=30 dyads (60 subjects); minimum 20 accurate + 20 inaccurate trials per subject per dyad

Analysis Cluster-based permutation test across time-frequency space; N=5000 permutations; cluster threshold $p<0.05$

Null condition Prediction fails if no cluster survives permutation threshold in the pre-response window

Prediction 4 — B index decreases after intranasal oxytocin (double-blind RCT)

Hypothesis Intranasal oxytocin (24 IU) reduces B-index vs. saline placebo

Design Double-blind crossover; 1-week washout; counterbalanced order

Confound controls Menstrual cycle phase; baseline B covariate; state anxiety (STAI-S)

Required N N=60 (paired t-test power: δ=0.25 SD, power=0.80, α=0.05)

Analysis Paired t-test; Cohen's d with 95% CI; Bayesian equivalent (BF_{10}) as sensitivity analysis

Null condition Prediction fails if |B_oxytocin − B_placebo| < 0.10

Prediction 5 — Acute mania shows supercritical EEG spectral radius

Hypothesis $\rho > 0.96$ on > 60% of 30-second EEG windows during acute mania (YMRS ≥ 20) vs. < 20% during euthymia (YMRS ≤ 6), within the same patients

Participants Bipolar I patients; N=40; EEG at hospitalisation (acute) and 6-month follow-up (euthymia)

Analysis Paired Wilcoxon signed-rank test; proportion of windows with $\rho>0.96$ per session; r effect size

Safety CAARMS weekly; YMRS at every EEG session; psychiatrist approval before enrolment

Null condition Prediction fails if proportions do not significantly differ ($p>0.05$)

Prediction 6 — Composite health index predicts clinical impairment

Hypothesis A composite of P, B, ρ, and η_r predicts GAF < 60 with AUC > 0.75 in ROC analysis

Design Cross-sectional; 30-min EEG + 8-min fMRI; GAF rated by clinician blinded to neurophysiology

Participants N=120: 40 healthy controls, 40 non-acute psychiatric outpatients, 40 acute-episode inpatients

Analysis Logistic regression; ROC analysis; AUC with 95% CI (DeLong method); Youden J threshold

Null condition Prediction fails if AUC lower 95% CI bound < 0.65

SECTION 4

Validated Analysis Function Library

Python pseudocode with complete input/output contracts

The following functions provide the analysis backbone for testing the predictions above. Each is specified with: input types, output types, algorithm reference, and stated failure mode. Full implementations with unit tests should be published in a public repository (GitHub + Zenodo DOI) before any manuscript submission.

Dependencies MNE-Python (≥1.7), scikit-learn (≥1.4), scipy (≥1.13), statsmodels (≥0.14)

Hardware Standard workstation: 16 GB RAM, 8-core CPU sufficient for EEG analyses

Signal Processing

F1. compute_mismatch_negativity(epochs, deviant_ids, standard_ids)

```
# Input:  MNE Epochs object; lists of event IDs
# Output: dict {amplitude_uV: float, latency_ms: float, SNR: float}
# Method: grand-average deviant ERP minus standard ERP; peak in 100–250ms window
# Fails:  ValueError if fewer than 40 deviant trials available
# Source: Näätänen et al. (2007) Clin Neurophysiol 118:2544
```

F2. source_localize_lcmv(raw, forward, noise_cov, data_cov)

```
# Input:  MNE Raw EEG; BEM forward model; noise and data covariance matrices
# Output: MNE SourceEstimate (source time series)
# Method: LCMV beamformer; unit-noise-gain normalisation
# Fails:  warns if SNR < 3 dB; raises if any channel impedance > 20 kΩ
# Source: Van Veen et al. (1997) IEEE Trans Biomed Eng 44:867
```

F3. compute_plv_surrogate_corrected(sig_a, sig_b, freq_band, n_surr=200)

```
# Input:  source-localised EEG arrays [T,]; frequency band (Hz_low, Hz_high)
# Output: dict {PLV_raw: float, PLV_corrected: float, p_value: float}
# Method: Hilbert transform → phase; subtract mean of 200 phase-randomised
surrogates
# Fails:  raises if n_surr < 200 or signal length < 500 ms
# Source: Lachaux et al. (1999) Hum Brain Mapp 8:194
```

F4. estimate_spectral_radius_mvar(eeg_data, max_order=20)

```
# Input:  ndarray [channels × timepoints]
# Output: dict {rho: float, model_order: int, stationarity_passed: bool}
# Method: ADF test per channel; AIC model selection; companion matrix eigenvalues
# Fails:  StationarityError if ADF fails on > 20% of channels
# Source: Kaminski & Blinowska (1991) Biol Cybern 65:203
```

F5. compute_mutual_information_ksg(X, Y, k=3)

```
# Input:  arrays X [n,], Y [n,]; int k (nearest neighbours)
# Output: dict {MI_bits: float, MI_se: float, p_value: float}
```

```
# Method: KSG k-NN estimator; bias-correct via 200 time-shuffled nulls
# Fails:  raises if n < 100
# Source: Kraskov et al. (2004) Phys Rev E 69:066138
```

Statistical Functions

F6. power_analysis_n(effect_type, effect_value, power=0.80, alpha=0.05)

```
# Input:  str effect_type ('d','r','f2'); float effect_value
# Output: int minimum_n_per_group
# Method: statsmodels.stats.power; returns ceil(n) + 10% buffer
# REQUIRED: call BEFORE data collection; log result in pre-registration document
```

F7. bootstrap_ci(data, stat_fn, n=10000, ci=0.95)

```
# Input:  array; callable statistic function; int iterations
# Output: dict {estimate: float, ci_low: float, ci_high: float, se: float}
# Method: percentile bootstrap; BCa correction if |skewness| > 0.5
# Fails:  raises if sample size < 30
```

F8. granger_causality_test(source, target, max_lag=10)

```
# Input:  arrays source [T,], target [T,]; int max_lag
# Output: dict {F_stat: float, p_value: float, optimal_lag: int, GC: float}
# Method: VAR model; AIC lag selection; F-test on restricted vs. unrestricted model
# Prerequisite: ADF stationarity test must pass on both series
```

F9. roc_analysis(scores, labels, n_bootstrap=2000)

```
# Input:  array continuous scores; binary labels
# Output: dict {AUC: float, AUC_ci: tuple, threshold_youden: float, sensitivity:
float, specificity: float}
# Method: sklearn.metrics.roc_auc_score; 95% CI via DeLong or bootstrap
```

Clinical Monitoring

F10. screen_participant(participant_data, study_type)

```
# Input:  dict {age, diagnoses, BSS_score, PANSS_P_scores, BP_systolic}
# Input:  str study_type ('healthy_eeg' | 'clinical_eeg' | 'stimulant_challenge')
# Output: dict {eligible: bool, exclusions: list[str]}
# Hard stop: BSS > 0 triggers HardExclusionError — must be logged and referred
```

F11. monitor_session_safety(vitals, mood_scores)

```
# Input:  dict vitals {BP_sys, BP_dia, HR_bpm, SpO2_pct}
# Input:  dict mood_scores {CAARMS_total, YMRS_total}
# Output: dict {continue: bool, flag: str, required_action: str}
# Stop conditions: BP_sys > 140, HR > 100, SpO2 < 95%, CAARMS increase > 5 pts
# Logs all events to encrypted session file regardless of continue decision
```

SECTION 5

Mandatory Ethical Safeguards

APA-compliant protocol elements for human subjects research

[MANDATORY] All research using this framework requires ethics board approval (IRB in the US; REC in the UK; equivalent internationally). These safeguards are preconditions for ethical research, not optional additions.

5.1 Universal Safeguards — All Studies

√ Written informed consent: timestamped; includes explicit statement that no paranormal claims are made and results are not diagnostic

√ Self-diagnosis warning prominently displayed before and after each session

√ Beck Scale for Suicidal Ideation (BSS) before and after each session; score > 0 = immediate clinical referral and session termination

√ All null results must be registered and published; suppression of negative results is an ethical violation in this context

√ Data retention maximum 5 years; AES-256 encryption; k-anonymisation before sharing

√ Statistical analysis plan fully specified in pre-registration before data collection; deviations documented post-hoc

5.2 Stimulant Challenge Studies — Additional Requirements

⚠️ **Stimulant studies require physician oversight and on-site medical emergency protocol.**

√ Participants: healthy adults 18–45 only; no personal or first-degree family history of psychosis, mania, or bipolar disorder

√ Doses: FDA-approved therapeutic maximum only (amphetamine ≤ 20 mg single oral dose); no repeat dosing within 7 days

√ Continuous monitoring: BP, HR, SpO2, ECG (QTc interval); automated stop if BP > 140/90 OR HR > 100 bpm

√ PANSS-P items at baseline; any item ≥ 4 = hard exclusion; do not enrol

⚠️ **Stimulant psychosis is a medical emergency — not an 'altered state of interest.' Protocols for emergency psychiatric consultation must be in place before study begins.**

5.3 Psychiatric Population Studies — Additional Requirements

⚠️ **BPD, bipolar, schizophrenia-spectrum, and PTSD studies require: clinical oversight, crisis protocol, and sessions ≤ 60 minutes.**

√ Licensed clinician present or immediately reachable for all sessions

√ BPD: DES (Dissociative Experiences Scale) before each session; DES > 30 = low-arousal protocol only

√ Bipolar: YMRS at each session; YMRS > 15 = EEG only; no stimulant challenge; no high-arousal induction

√ Schizophrenia: use 'altered temporal perception' not 'precognition'; never romanticise positive symptoms

√ PTSD: written safety plan provided; stabilisation phase required before any exploratory protocols

√ Medication adherence assessed at each session; non-adherence documented and psychiatrist notified

5.4 Required Consent Language Elements

Consent Element	Required Language
Purpose	This research studies computational measures of brain activity. It does not study paranormal or psychic phenomena.
Paranormal disclaimer	No claim is made or implied that any result indicates supernatural ability. All phenomena studied have known physical mechanisms.
Clinical disclaimer	Results cannot be used for diagnosis, clinical assessment, or treatment decisions.
Voluntary withdrawal	You may stop at any time without penalty. Data will be deleted on request before analysis begins.
Data sharing	De-identified data will be shared in a public repository. Your identity cannot be recovered from shared data.
Risk statement	Risks: mild fatigue from EEG sessions. For stimulant studies: cardiovascular effects are continuously monitored by clinical staff.

5.5 APA Ethical Principles Compliance

Beneficence & Nonmaleficence (3.04) Minimise harm; clinical monitoring and stop rules operationalise this requirement

Fidelity & Responsibility (1.01) Do not misrepresent this framework as established science; honest classification maintained throughout

Integrity (2.04) Report all results including null; suppress nothing

Justice (7.02) Do not exploit vulnerable populations; enhanced protections for psychiatric participants throughout

Respect for Rights (3.10) Informed consent documented and timestamped; voluntary participation made explicit

SECTION 6

Physics and Logic Compliance

Constraints that prevent pseudoscientific drift

Every version of the theoretical framework that proposes a causal mechanism must respect the following constraints. Any equation or claim that violates these is wrong — not merely speculative — and must be corrected or removed.

Constraint	Statement	Implication for This Framework
Causality	t_effect > t_cause in all classical systems	'Precognition' = predictive extrapolation from past patterns, not retrocausation
Conservation of energy	Metabolic cost applies to all neural processing	Every proposed coherence state has a measurable ATP cost
Shannon capacity	$C = B \cdot \log_2(1+SNR)$ bounds all information transfer	Social 'bandwidth' is bounded; the channel must be specified and measured
Second law	$\Delta S_total \geq 0$	Information exchange has irreversible thermodynamic costs
Neural transmission locality	Conduction velocity ≈ 0.5–100 m/s	No instantaneous inter-brain link; all synchrony has finite propagation delay
Standard model of physics	All forces are electromagnetic, weak, strong, or gravitational	No unknown force is invoked; all mechanisms reducible to known interactions
Falsifiability	Every scientific claim specifies conditions under which it is false	All equations include null conditions — mandatory, not optional
Occam's razor	Prefer simpler explanation absent evidence for complexity	If cue-reading explains a result, no additional mechanism is invoked
Biological realism	Mechanisms must map to known cellular and network biology	'Holographic' claims require a dendritic/synaptic mechanism to be more than analogy
Reproducibility	Results must replicate in independent laboratories	Single-laboratory theoretical synthesis is a hypothesis; it is not a finding

[QUANTUM ANALOGY WARNING] The framework uses quantum-mechanical language analogically. None of this implies quantum mechanisms in the brain. Neural dynamics are classical. Tegmark (2000) established that microtubule quantum coherence decoheres in ~10^{-13} s — far below the ~10 ms timescale of neural dynamics. Any future version claiming a quantum mechanism must: (1) identify the specific quantum system, (2) demonstrate decoherence time exceeds the relevant neural timescale, and (3) propose a prediction that distinguishes quantum from classical explanation.

CONCLUSION

What Is This, If Not Science?

The honest accounting

This companion validation protocol — like the theoretical framework it serves — must be evaluated honestly.

What the Validation Protocol Is

√ A rigorous, dimensionally verified set of measurement specifications for the framework's three axes
√ A collection of pre-registerable experimental designs that would, if executed, begin to test the framework's claims
√ A set of ethical safeguards enabling responsible human-subjects research on these questions
√ A library of validated analysis functions drawn from published methods in the relevant literature

What the Validation Protocol Is Not

⚠️ **Not data. Not results. Not evidence. Not a scientific finding.**
⚠️ **Not a clinical instrument — nothing here is validated for clinical use**
⚠️ **Not peer-reviewed — these protocols have not been evaluated by independent domain experts**
⚠️ **Not pre-registered — none of the predictions have yet been registered before data collection**

The framework and this protocol together constitute what might be called a proto-scientific edifice: a structure built with scientific materials, in a scientific style, not yet inhabited by scientific data.

THE HONEST CONCLUSION:

This body of work is an exercise in computational philosophy and speculative world-building. It simulates the aesthetic of scientific discovery — numbered equations, Greek letters, acronyms, appeals to peer review — without performing the actual labour of scientific discovery: controlled experiments, real participants, pre-registered analyses, peer-reviewed publication, and independent replication.

What it accomplishes is different, and genuinely valuable: it generates a large family of precisely specified hypotheses about the neural basis of altered social cognition. Those hypotheses are interesting, internally consistent, and connect to real empirical literature. Several have moderate-to-high plausibility given existing findings.

A precisely specified, falsifiable hypothesis is the necessary precondition for scientific work. This framework has produced many of them. The next step — the only step that can change its classification — is for someone to pick one, pre-register it, collect the data, and publish the result regardless of what it shows.

The protocol is the map to the laboratory. The laboratory has not yet been entered. This is how to enter it.